Question Paper: Machine Learning

Machine Learning Question Paper

Instructions:

* The question paper consists of three sections: Multiple Choice Questions (MCQs), Short Questions, and Long/Descriptive Questions.

* Attempt all questions in the given time frame.

* Use a blue or black pen to write your answers.

Section A: Multiple Choice Questions (1-8)

1. Which of the following is a type of machine learning algorithm?

A) Linear Programming

B) Decision Trees

C) Web Development

D) Data Mining

2. What is the primary goal of a machine learning model?

A) To minimize error

B) To maximize accuracy

C) To reduce complexity

D) To increase data size

3. Which of the following is a supervised learning algorithm?

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A) K-Means Clustering

B) Linear Regression

C) Decision Trees

D) Apriori Algorithm

4. What is the term for the process of training a machine learning model on a large dataset?

A) Overfitting

B) Underfitting

C) Regularization

D) Hyperparameter Tuning

5. Which of the following is a type of neural network?

A) Decision Tree

B) Support Vector Machine

C) Convolutional Neural Network

D) Linear Regression

6. What is the purpose of cross-validation in machine learning?

A) To evaluate model performance

B) To select the best features

C) To tune hyperparameters

D) To reduce overfitting

7. Which of the following is a clustering algorithm?

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A) Linear Regression

B) Decision Trees

C) K-Means Clustering

D) Support Vector Machine

8. What is the term for the process of selecting the most relevant features for a machine learning model?

A) Feature Extraction

B) Feature Selection

C) Dimensionality Reduction

D) Data Preprocessing

Section B: Short Questions (9-12)

9. What is the difference between supervised and unsupervised learning? (3 marks)

10. Explain the concept of overfitting in machine learning. (4 marks)

11. Describe the steps involved in building a machine learning model. (5 marks)

12. What is the role of regularization in machine learning? (3 marks)

Section C: Long/Descriptive Questions (13-14)

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13. Describe the architecture of a convolutional neural network (CNN) and its applications in image classification. (8 marks)

14. Explain the concept of reinforcement learning and its difference from supervised and unsupervised learning. Provide an example of a reinforcement learning algorithm. (10 marks)

Answer Key

1. B) Decision Trees
2. B) To maximize accuracy
3. B) Linear Regression
4. D) Hyperparameter Tuning
5. C) Convolutional Neural Network
6. A) To evaluate model performance
7. C) K-Means Clustering
8. B) Feature Selection
9. Answer should include: Supervised learning involves training a model on labeled data, while unsupervised learning involves training a model on unlabeled data.
10. Answer should include: Overfitting occurs when a model is too complex and performs well on the training data but poorly on new, unseen data.
11. Answer should include: Steps involved in building a machine learning model include data preprocessing, feature selection, model selection, training, and evaluation.
12. Answer should include: Regularization is a technique used to prevent overfitting by adding a penalty term to the loss function.

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13. Answer should include: A CNN consists of convolutional layers, pooling layers, and fully connected layers. Applications include image classification, object detection, and image segmentation.

14. Answer should include: Reinforcement learning involves training an agent to take actions in an environment to maximize a reward signal. Difference from supervised and unsupervised learning: reinforcement learning involves trial and error, while supervised and unsupervised learning involve learning from data. Example: Q-learning algorithm.